

PRINCE CHAUHAN

SENIOR QUANTITATIVE ANALYST & RESEARCHER

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SEBI Registered Research Analyst

PROFESSIONAL SUMMARY

Derivatives research professional with 9+ years of experience spanning options trading, market analysis, and systematic strategy development. Focused on Indian equity derivatives and US index options, with Python-based research covering volatility modeling, statistical arbitrage, and execution analysis. Experienced in strategy validation, portfolio risk monitoring, and research automation.

PROFESSIONAL EXPERIENCE

SMC Global Securities Ltd. - Senior Quantitative Analyst & Researcher

Dec 2024 - Present

- **Alpha Research Infrastructure:** Architected event-driven simulation pipelines and point-in-time research infrastructure on tick-level and 1-minute historical bar feeds across US and Indian index derivatives (SPX, VIX, Nifty, Bank Nifty).
- **US Index Derivatives:** Engineered automated execution pipelines and volatility surface modeling for short-dated US index options (SPX/SPXW) via Interactive Brokers (IBKR API), systematically capturing short-horizon IV skew mispricings with dynamic delta rebalancing.
- **Statistical Arbitrage:** Formulated Bank Nifty constituent pair and basket models applying Johansen cointegration rank tests, dynamic rolling beta hedging, and spread mean-reversion half-life estimation to systematically capture relative-value opportunities.
- **Chronological Walk-Forward Backtesting:** Evaluated systematic volatility strategies across Nifty and Bank Nifty index options over a 7-year sample (2018-2024); achieved an annualized out-of-sample Sharpe of 2.5+ and profit factor of 2.1+ after deducting 0.50% round-trip option premium turnover friction (covering STT, exchange fees, and bid-ask crossing slippage).
- **Execution Architecture & Telemetry:** Built pre/post-trade Transaction Cost Analysis (TCA) and live telemetry over an asynchronous pipeline (Python Asyncio, WebSockets, mmap IPC), reducing execution drag by 24% and eliminating broker order-state desynchronization and duplicate executions via atomic client order IDs.
- **Desk Mentorship & Risk Alignment:** Mentored 3 junior quantitative researchers and traders in statistical validation methods and execution best practices; partnered with Portfolio Managers on strategy scalability, real-time portfolio Greeks, dealer GEX mapping, parametric VaR, and SPAN margin caps.

Smart Bulls Capital - Research Analyst, Derivatives & Quantitative Research

Mar 2017 - Nov 2024

- **Options Volatility Books:** Researched, structured, and traded systematic multi-leg options volatility strategies (Iron Condors, Butterflies, Calendar Spreads) with dynamic delta/vega rebalancing guided by real-time IV surface spline smoothing.
- **Mathematical Root-Finding Solvers:** Computed implied volatility by numerically inverting Black-Scholes option prices using Newton-Raphson and Brent root-finding algorithms; mapped dealer Gamma Exposure (GEX) profiles to quantify institutional pin risk.
- **Portfolio Construction & Allocation:** Implemented Risk Parity and Mean-Variance capital allocation models across multi-strike options books; managed daily P&L attribution, SPAN margin utilization, and drawdown limits within desk risk budgets.
- **Fundamental Balance-Sheet Gating:** Conducted forensic financial statement audits, DCF valuation, and working-capital variance analysis across banking and industrial equities to gate capital deployment to quantitative models.

Earlier Corporate Experience - Corporate Finance & Audit Roles

2014 - 2017

- **Corporate Treasury & Audit:** Executed treasury operations and working-capital forecasting at Indian Oil Corporation Ltd. (IOCL, 2016-2017); conducted statutory audits and corporate tax structuring at CA Matta & Associates (2014-2016).

SELECTED INDEPENDENT QUANTITATIVE RESEARCH (OPEN SOURCE)

Hosted at: github.com/prince-chauhan-pc/quant-research-portfolio

Point-in-Time Backtesting Framework & Invariant Gate (Python)

2024

- **Research Objective:** Enforce point-in-time data alignment and realistic transaction friction in high-turnover options strategy research.
- **Methodology & Validation:** Implemented rolling walk-forward optimization partitioned into 70% in-sample parameter calibration, 20% out-of-sample cross-validation, and 10% held-out test windows to prevent data leakage, paired with 1,000-path Monte Carlo drawdown stress testing.
- **Audit Guardrails:** Engineered an automated data-integrity test suite enforcing Rules L1-L17 across 22 boundary checks, pre-deducting 0.50% round-trip premium friction and exchange SPAN margins.

Arbitrage-Free SVI Volatility Surface & GEX Profiling

2024

- **Research Objective:** Eliminate calendar and butterfly arbitrage from market implied volatility quotes for robust cross-strike option pricing.
- **Methodology:** Calibrated daily implied volatility surfaces using SVI parameterization with non-linear optimization; enforced butterfly convexity and calendar-spread non-arbitrage constraints.
- **Practical Outcome:** Stabilized cross-strike smile interpolation across illiquid strikes and generated real-time dealer GEX sensitivity curves to map institutional pin-risk zones.

Bank Nifty Basket Cointegration & Statistical Arbitrage

2024

- **Research Objective:** Systematically exploit stationary mean-reverting spreads among top banking constituent equities.
- **Methodology & Risk:** Applied Johansen multivariate cointegration tests to identify stationary linear combinations; estimated spread mean-reversion half-life to calibrate entry/exit thresholds and holding periods, maintaining delta neutrality via rolling OLS beta weights.

Microstructure Execution Simulator & TCA Engine

2024

- **Research Objective:** Accurately quantify market crossing costs and adverse selection for algorithmic limit orders.
- **Methodology:** Modeled Limit Order Book (LOB) queue priority, non-linear market impact, and fill probability across market regimes classified by Parkinson realized volatility.

TECHNICAL & QUANTITATIVE COMPETENCIES

- **Quantitative Research:** Statistical Arbitrage, Johansen Cointegration, Dynamic Beta Hedging, SVI Volatility Surface Fitting, Parkinson Realized Volatility, Market Regime Classification (K-Means), Walk-Forward Optimization, Monte Carlo Simulation.
- **Derivatives & Risk:** Black-Scholes Pricing, Portfolio Greeks (Delta, Gamma, Vega, Theta), Dealer GEX Profiling, Parametric & Historical VaR, SPAN Cross-Margin Optimization, Risk Parity, Mean-Variance Allocation.
- **Programming & Data Systems:** Python 3.11+ (Polars, DuckDB, NumPy, SciPy, Statsmodels, Scikit-Learn, Asyncio), SQL (DuckDB, PostgreSQL), WebSockets, Memory-Mapped IPC (mmap), Apache Parquet, Interactive Brokers API (IBKR / ib_insync), DhanHQ SDK, Git, Linux/Bash.
- **Markets & Microstructure:** Indian Equity Derivatives (NSE Nifty, Bank Nifty, Sensex), US Equity Derivatives (CBOE SPX, VIX), Pre/Post-Trade TCA, Limit Order Book Queue Dynamics, Smart Order Slicing.

EDUCATION & REGULATORY CREDENTIALS

- **Bachelor of Commerce (Honours):** Kirori Mal College, University of Delhi (2011 - 2014) | First Division (71% Marks)
- **Chartered Accountancy (CA Final):** Institute of Chartered Accountants of India (ICAI, 2018) | Cleared Group 1 (78% Marks in Financial Reporting & Auditing)
- **Regulatory Licenses:** SEBI Registered Research Analyst | NISM Series-XV